

Radioprotection by plant products: present status and future prospects.

The development of radioprotective agents has been the subject of intense research in view of their potential for use within a radiation environment, such as space exploration, radiotherapy and even nuclear war. However, no ideal, safe synthetic radioprotectors are available to date, so the search for alternative sources, including plants, has been on going for several decades. In Ayurveda, the traditional Indian system of medicine, several plants have been used to treat free radical-mediated ailments and, therefore, it is logical to expect that such plants may also render some protection against radiation damage. A systematic screening approach can provide leads to identifying potential new candidate drugs from plant sources, for mitigation of radiation injury. This article reviews some of the most promising plants, and their bioactive principles, that are widely used in traditional systems of medicine, and which have rendered significant radioprotection in both in vitro and in vivo model systems. Plants and their constituents with pharmacological activities that may be relevant to amelioration of radiation-mediated damage, including antiemetic, antiinflammatory, antioxidant, cell proliferative, wound healing and haemopoietic stimulatories are also discussed. Copyright (c) 2005 John Wiley & Sons, Ltd.